Inverse Kinematic Solution of 6-DOF Robot-Arm Based on Dual Quaternions and Axis Invariant Methods

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ABSTRACT

Solving inverse kinematics for a 6R robot using traditional methods is prone to singularities. This paper proposes a closed-form algebraic and geometric solution based on dual quaternions and axis invariants, achieving extremely high computational efficiency and singularity avoidance.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Validates the analytical inverse kinematics approach implemented on the ESP32 S3 for real-time Cartesian control.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Abubaker Ahmed provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.